

Ex. No. : 01
Register No.: 231701025

Date: 17/02/26
Name: Kamalesh R

A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE, BIVARIATE AND MULTIVARIATE REGRESION

Aim:

Develop a Python program to implement **univariate, bivariate, and multivariate regression** models and analyze the relationship between dependent and independent variables.

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import kagglehub
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Import Libraries

path = kagglehub.dataset_download("imdevskp/corona-virus-report")
df = pd.read_csv(os.path.join(path, "country_wise_latest.csv"))

print("Dataset Shape:", df.shape)
print(df.head())

# Univariate Analysis

print(df['Confirmed'].describe())
```

```
plt.hist(df['Confirmed'], bins=30)
plt.xlabel("Confirmed Cases")
plt.ylabel("Frequency")
plt.title("Univariate Analysis - Confirmed Cases")
plt.show()
```

Bivariate Analysis

```
plt.scatter(df['Confirmed'], df['Deaths'])
plt.xlabel("Confirmed Cases")
plt.ylabel("Deaths")
plt.title("Bivariate Analysis - Confirmed vs Deaths")
plt.show()
```

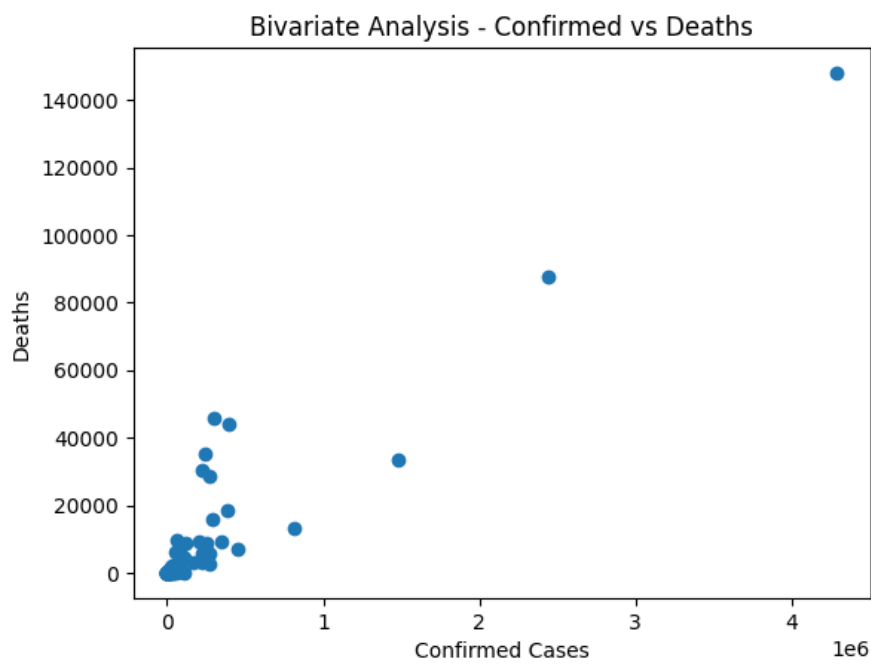
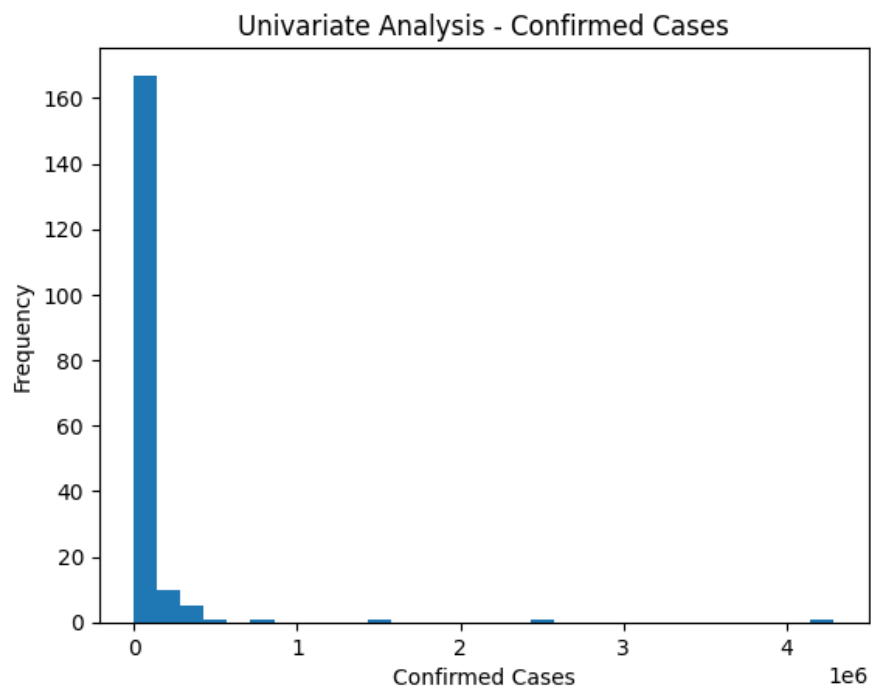
```
print("Correlation:", df['Confirmed'].corr(df['Deaths']))
```

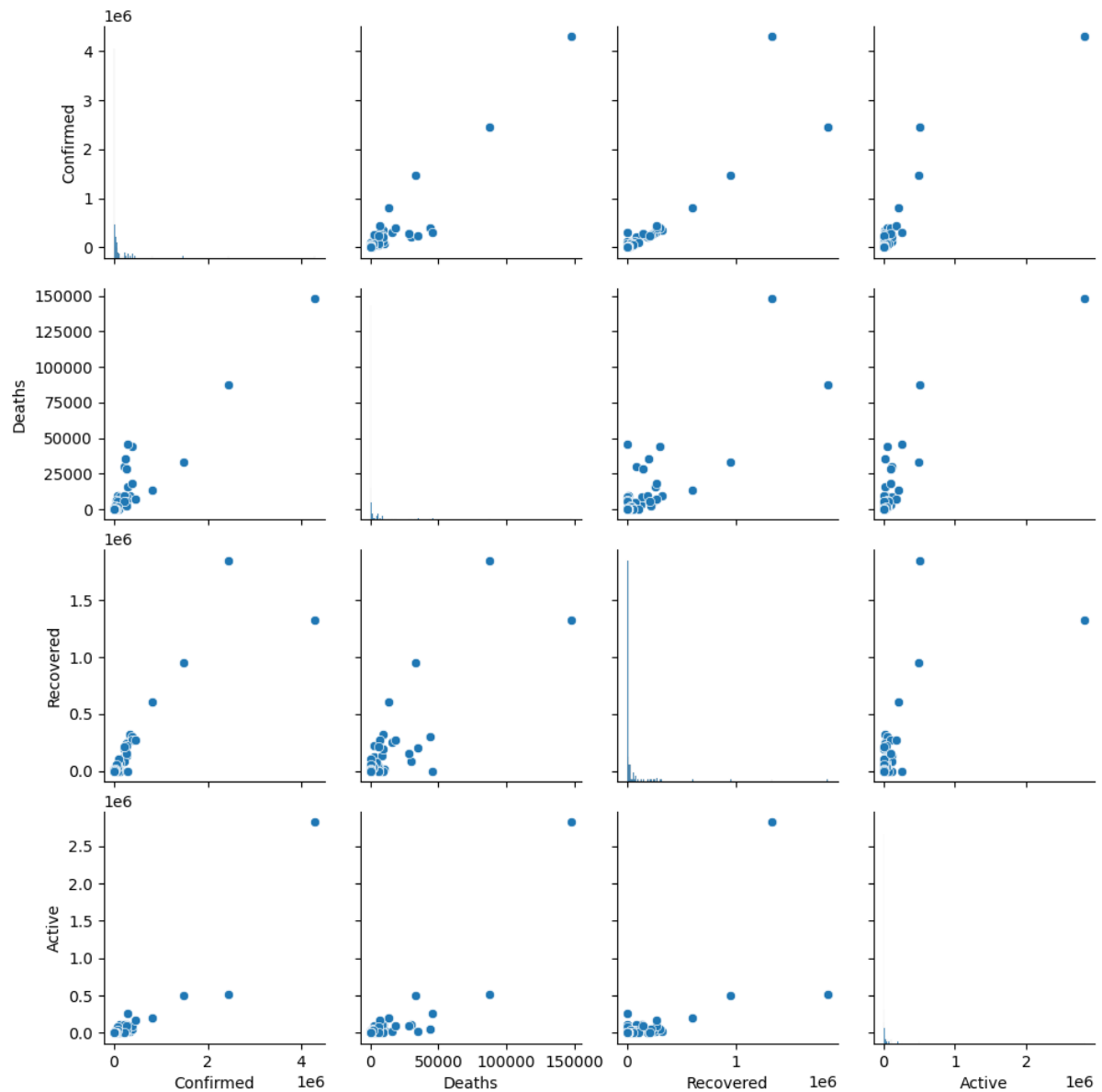
Multivariate Analysis

```
selected_columns = df[['Confirmed','Deaths','Recovered','Active']]
corr_matrix = selected_columns.corr()
print(corr_matrix)
sns.heatmap(corr_matrix, annot=True)
plt.title("Correlation Heatmap")
plt.show()
sns.pairplot(selected_columns)
plt.show()
```

Output

:





#Result:

The Python program for univariate, bivariate, and multivariate regression was successfully implemented using the Iris.csv dataset in Jupyter Notebook, and the regression models were able to analyze and predict relationships between variables accurately.